

On the effect of the magnetic compressibility in microtearing turbulence

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Introduction

- Microtearing (MT) mode turbulence, driven by the electron temperature gradient, leads to electron anomalous thermal transport in magnetic confinement devices, as demonstrated by a number of gyrokinetic simulations dedicated to specific experimental devices [e.g., Guttenfelder et al., 2011; Jian et al., 2019].
- The destabilization mechanism of MT mode is the time-dependent thermal force due to the electron temperature gradient and the velocity-dependent collision effect [Hazeltine et al., 1975, Hassam, 1980]. This mechanism has the drift-wave-like physical picture [Yagyu and Numata, 2023].
- The thermal transport level for electrons is theoretically predicted to scale as $|B_{\perp}|/B_0 \sim \rho_e/L_{T_{0e}}$, which is derived from a drift-kinetic model in collisional plasmas [Drake et al., 1980]. This prediction has been demonstrated by gyrokinetic simulations using experimental parameters [Wong et al., 2007; Doerk et al., 2011].

In our previous study [Yagyu and Numata, 2023], the preliminary nonlinear simulation has indicated that the energy via the electron temperature fluctuation may be dominant in microtearing turbulence.

Question

Does **the magnetic compressibility**, which is balanced by the perpendicular pressure fluctuation, **affect the transport level in microtearing turbulence**?

Summary

- We have performed gyrokinetic simulations of MT turbulence using gyrokinetic code GS2.
- MT modes become unstable due to the time-dependent thermal force, even in the absence of magnetic shear. The shear only plays a roll in stabilizing the MT mode.
- In nonlinear phase, energy owing to the electron temperature fluctuation is dominant, balancing that of the magnetic compressibility.
- The magnetic compressibility may enhance the level of thermal transport for electrons in MT turbulence. Note that zonal component is strong in our simulation result.
- It is shown theoretically that the effect of the magnetic compressibility modifies the fluctuating temperature profile through the perturbed diamagnetic flow, and can thereby affect the saturation level of MT turbulence, especially in higher-beta devices!
- We need to verify our theoretical result against turbulence-dominated case using gyrokinetic simulations.

Acknowledgments

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Theoretical and numerical analyses in 2D slab

Saturation mechanism and the effect of the magnetic compressibility in microtearing turbulence

We review the saturation mechanism of microtearing turbulence following Ref. [Drake et al., 1980]. The electrostatic potential ϕ and magnetic compressibility $\delta B_{\parallel}$ are ignored, and magnetic shear is also neglected, in Drake's analyses.

In the system in which the perturbation is sufficiently developed, the nonlinear term becomes dominant. Then, the reduced drift-kinetic equation for electrons is given by

$$\frac{\partial h_e}{\partial t} \sim \frac{v_{\parallel}}{B_0} \{A_{\parallel}, h_e\} \sim v_{\parallel} \frac{B_{\perp}}{B_0} \cdot \nabla_{\perp} h_e \quad (1) \quad \begin{array}{l} h_e : \text{non-Boltzmann part of dist. func.} \\ \text{obeying the drift-kinetic eqn.} \\ \{a, b\} = \partial_x a \partial_y b - \partial_y a \partial_x b : \text{Poisson bracket} \end{array}$$

Roughly estimate the sat. amp. of $|B_{\perp}| \longrightarrow \frac{|B_{\perp}|}{B_0} \sim \frac{\omega_{*e}}{k_y v_{th,e}} \sim \frac{\rho_e}{L_{T_{0e}}} \quad \frac{\partial}{\partial t} \rightarrow \omega \sim \omega_{*e}, \quad \nabla_{\perp} \rightarrow k_{\perp} \sim k_y, \quad v_{\parallel} \sim v_{th,e}$

The generalized Ohm's law, obtained by taking the velocity moment of eqn. (1), indicates that the temperature fluctuations profile is required to understand the nonlinear interaction.

$$\frac{\partial u_{\parallel,e}}{\partial t} \sim v_{th,e}^2 \frac{B_{\perp}}{B_0} \cdot \nabla_{\perp} \frac{\delta T_e}{T_{0e}}$$

The heat transport equation derived by [Hassam, 1980] is

$$\frac{3n_0}{2} \frac{\partial \delta T_e}{\partial t} + \dots = \kappa_{\parallel} \nabla_{\parallel}^2 \delta T_e - \frac{5n_0 T_{0e}}{2 q_e B_0} \nabla_{\perp} \cdot \left(\frac{\mathbf{B}}{B_0} \times \nabla_{\perp} \delta T_e \right) + \dots$$

$\equiv -\nabla_{\perp} \cdot \mathbf{q}_{\parallel} \quad \quad \quad \nabla_{\parallel} \equiv (\mathbf{B}/B_0) \cdot \nabla_{\perp}$
(if $\mathbf{B} \rightarrow \mathbf{B}_{\perp} \Rightarrow \nabla_{\perp} \cdot \mathbf{q}_{\text{dia}} = 0$)

The heat diffusion along the perturbed magnetic field cancels the temperature gradient, which is the driving force of turbulence, saturating MT turbulence.

If $\delta B_{\parallel} \neq 0$,

$$\mathbf{u}_{\text{dia}} \equiv (q_i/q_e) \mathbf{u}_{\text{dia},i} + \mathbf{u}_{\text{dia},e} \approx \mathbf{u}_{\text{dia},e} \propto \mathbf{B}_0 \times \nabla_{\perp} \delta P_{\perp,e}$$

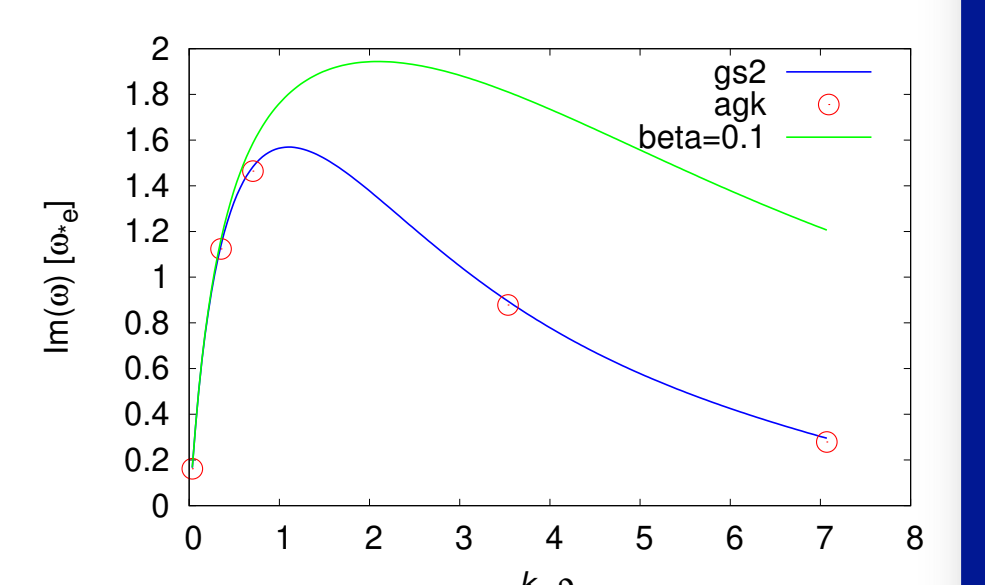
$$\nabla_{\perp} \cdot \mathbf{q}_{\text{dia}} = \frac{5n_0 T_{0e}}{2 q_e B_0} \left(\nabla_{\perp} \times \frac{\delta B_{\parallel}}{B_0} \hat{\mathbf{z}} \right) \cdot \nabla_{\perp} \delta T_e = \frac{5n_0 T_{0e}}{2} \frac{\beta_e}{q_e} \mathbf{u}_{\text{dia}} \cdot \nabla_{\perp} \frac{\delta T_e}{T_{0e}} \quad \beta_e \equiv \frac{2\mu_0 n_0 T_{0e}}{B_0^2}$$

Heat diffusion arises from the nonlinear interaction between the **perturbed diamagnetic flow** and the temperature fluctuation.

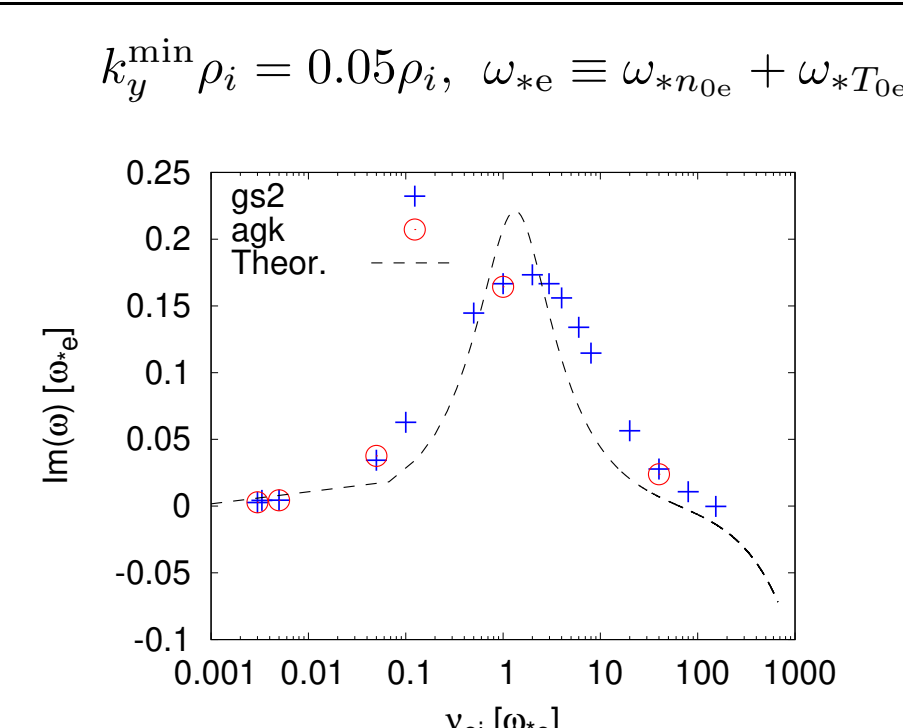
We investigate whether $\nabla_{\perp} \cdot \mathbf{q}_{\text{dia}}$ enhances $\nabla_{\perp} \cdot \mathbf{q}_{\parallel}$ using gyrokinetic simulations!

Simulation setup

- We perform gyrokinetic simulations using gyrokinetic code GS2.
- The magnetic shear is ignored as in [Drake et al., 1980]: $B_0 = B_0 \hat{\mathbf{z}}$ but fields $\phi, A_{\parallel}, \delta B_{\parallel}$ are included.
- Temperature and density gradients for electrons are given by $L_{n_{0e}}, L_{T_{0e}} \equiv -d \ln n_{0e}/dx, -d \ln T_{0e}/dx \quad \omega_{*e, n_{0e}, T_{0e}} \equiv \frac{k_y \rho_e}{2L_{n_{0e}, T_{0e}}} v_{th,e}$
- Collisions: Fokker-Planck collision operator.
- Physical Parameters: $\beta_e = 0.1, \quad \tau \equiv T_{0i}/T_{0e} = 0.5 \quad \sigma \equiv m_e/m_i = 1/1836, \quad \eta \equiv L_{n_{0e}}/L_{T_{0e}} = 5, \quad \nu_{ei}/\omega_{*e}(k_y^{\min}) = 3.$
- Nonlinear simulation setup
Numerical parameters: $(N_x, N_y, N_{\lambda}, N_E) = (128, 128, 16, 16)$
 $0.05 \rho_i \leq k_{x,y} \rho_i \leq 2.1 \rho_i$
- Hyper-viscosity parameters: $D_{\text{hyp}} = 10 \omega_{*e} (k_y = k_y^{\max}), \quad k_{x,y}^{\text{cut}} \approx 0.8 k_{x,y}^{\max}$



Linear simulation results



- Modes are unstable due to the time-dependent thermal force even in the absence of mag. shear.
- Growth rate depends non-monotonically on the collision frequency, like in MT with shear.
- Theor. curve agrees with the simulation results.

Theoretical solution derived from a drift-kinetic equation:

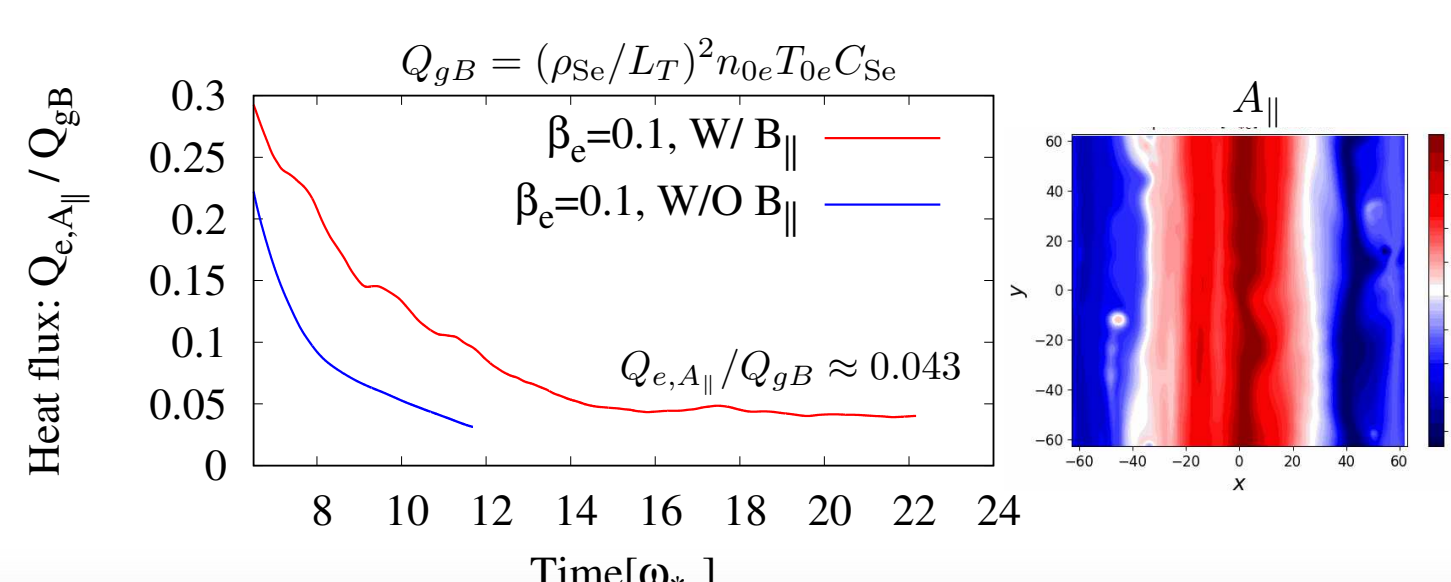
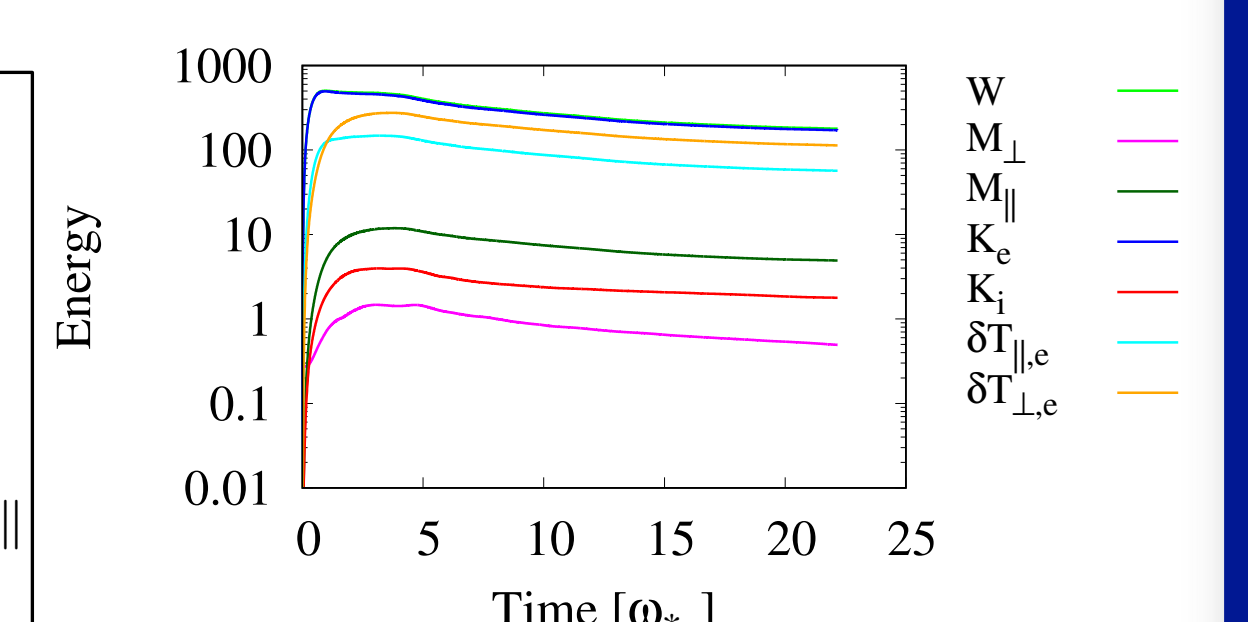
$$\alpha_{\eta} (k_y d_e)^2 = i \left[\frac{\omega}{\nu_{ei}} - \frac{\omega_{*n_{0e}}}{\nu_{ei}} (1 + (1 + \alpha) \eta_e) \right] - \alpha \alpha' \frac{\omega}{\nu_{ei}} \left(\frac{\omega}{\nu_{ei}} - \frac{\omega_{*n_{0e}}}{\nu_{ei}} (1 + \alpha' \eta_e) \right)$$

$\alpha, \alpha', \alpha'', \alpha_{\eta} : \text{Numerical factor}$

Nonlinear simulation results

- The generalized energy in gyrokinetic system:
 $W = \frac{1}{V} \int \left[\sum_s \int \frac{T_{0s} \delta f_s^2}{2 f_{0s}} d\mathbf{v} + \frac{|\nabla_{\perp} A_{\parallel}|^2}{2 \mu_0} + \frac{|\delta B_{\parallel}|^2}{2 \mu_0} \right] d\mathbf{r} = \sum_s K_s + M_{\perp} + M_{\parallel}$
- Electron particle energy is dominant and supported by $\delta T_{\parallel,e}, \delta T_{\perp,e}$.
- Energy associated with $\delta B_{\parallel}$ is higher than that of $A_{\parallel}$ and $\delta B_{\parallel}/B_0 \approx \beta_e \delta T_{\perp,e}/T_{0e}$ (Perp. Ampère's law).

$$\delta f_s = h_s - \frac{q_s \varphi}{T_{0s}}$$



- Heat flux along $B_{\perp}$ is $Q_{e,A_{\parallel}}/Q_{gB} \approx 0.043$, $Q_{e,EM}^{Doerk}/Q_{gB} \sim \mathcal{O}(0.1)$ see [Doerk et al., 2012]
- Heat flux without $\delta B_{\parallel}$ is likely to be smaller than the case with $\delta B_{\parallel}$.
- Zonal component is strong.

Confirm if the same results are obtained even when turbulence is dominant!